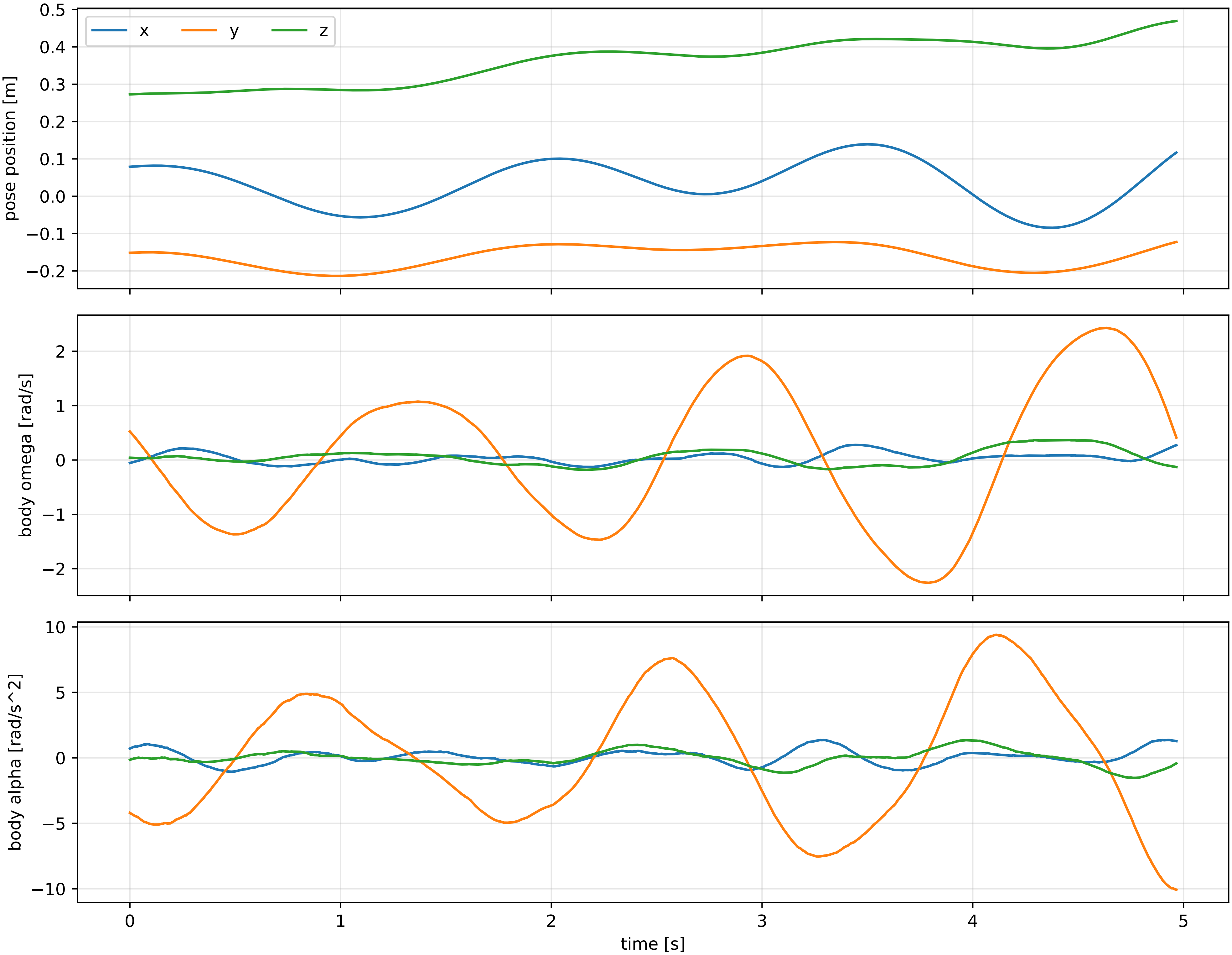
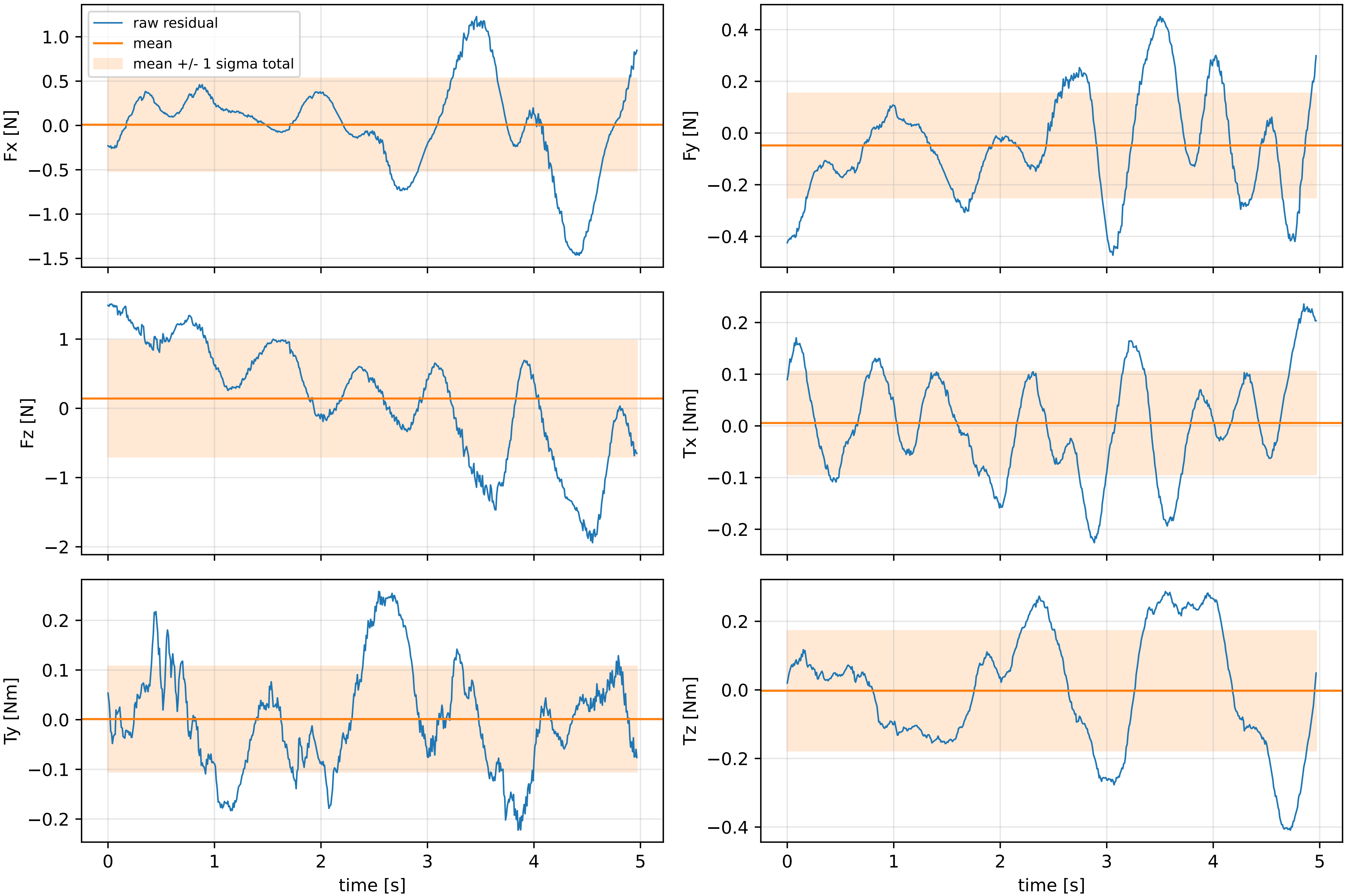
cog_y_nominal: SG trajectory and derived motion

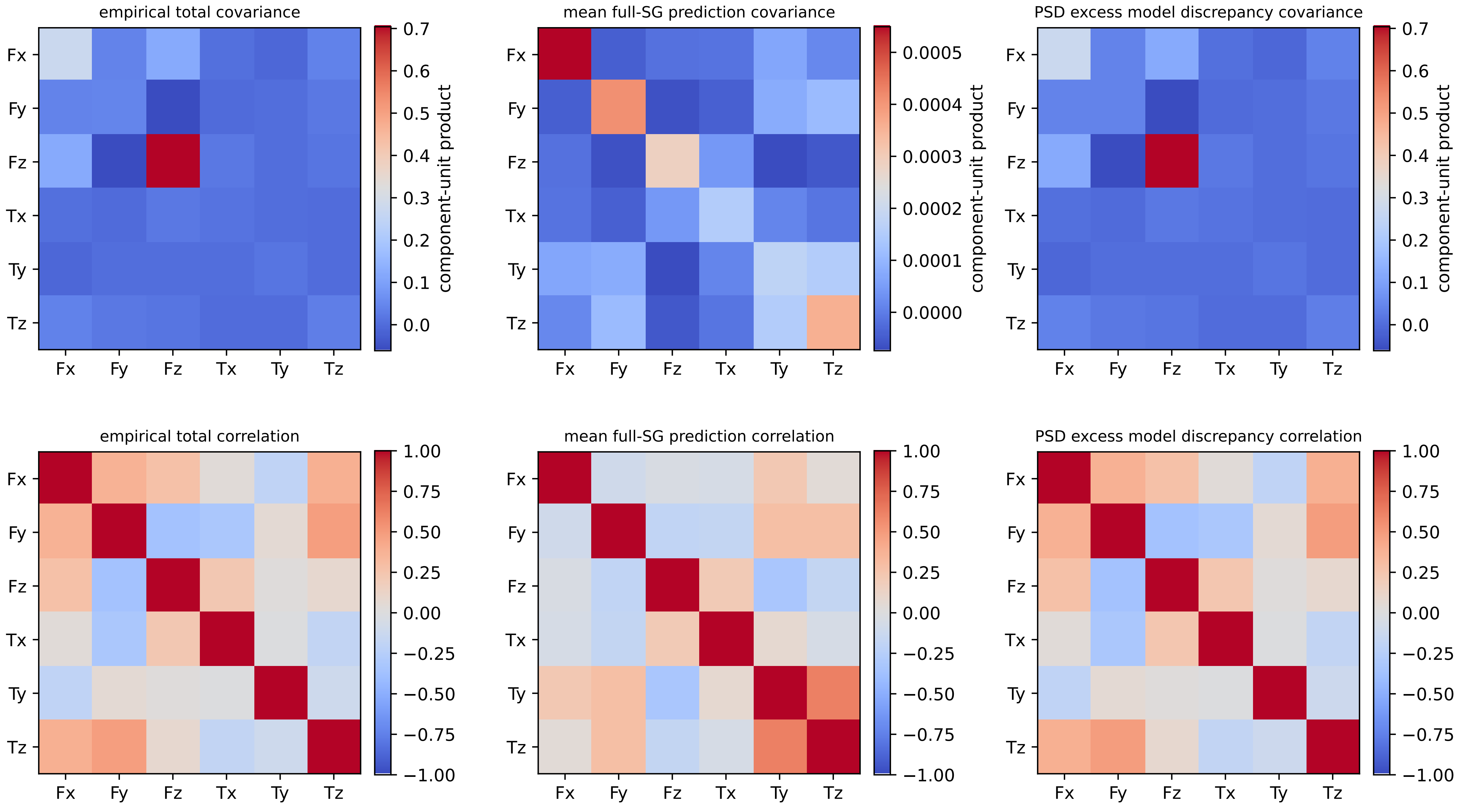


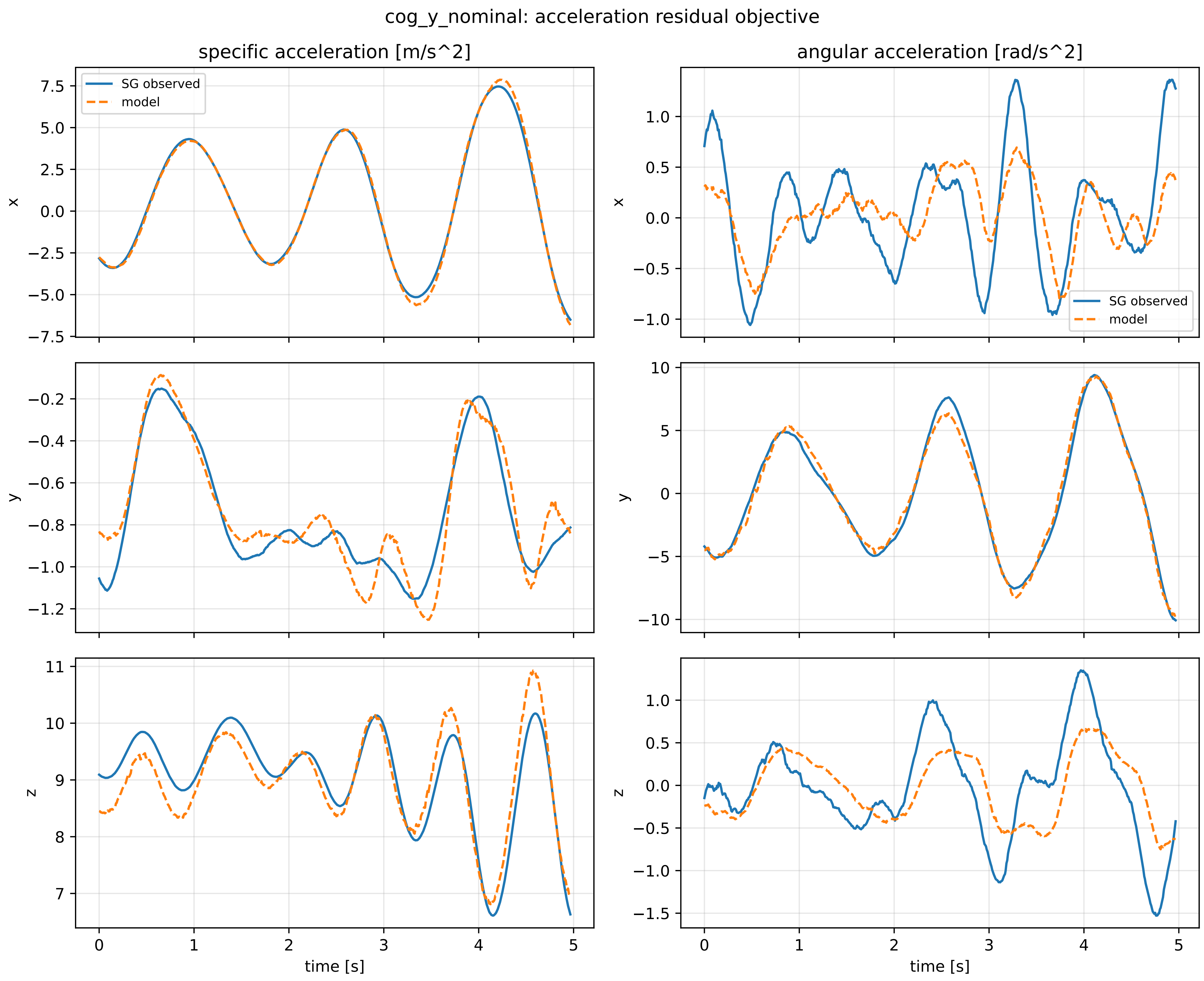
cog_y_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



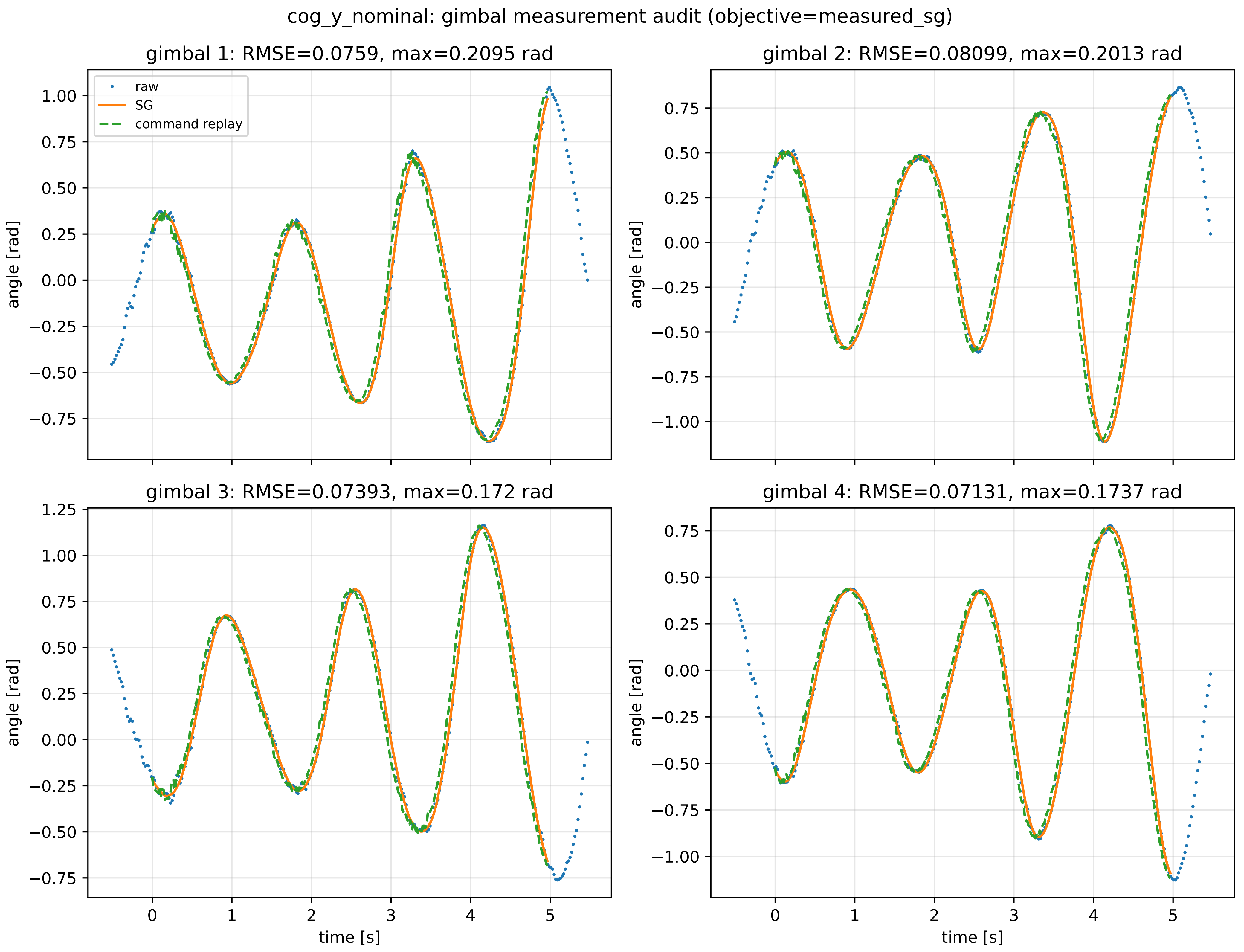
vector RMS about zero: force 1.0191 N, torque 0.22692 Nm; component std about mean: force [0.5226 0.2012 0.8392], torque [0.0995 0.1059 0.1744]

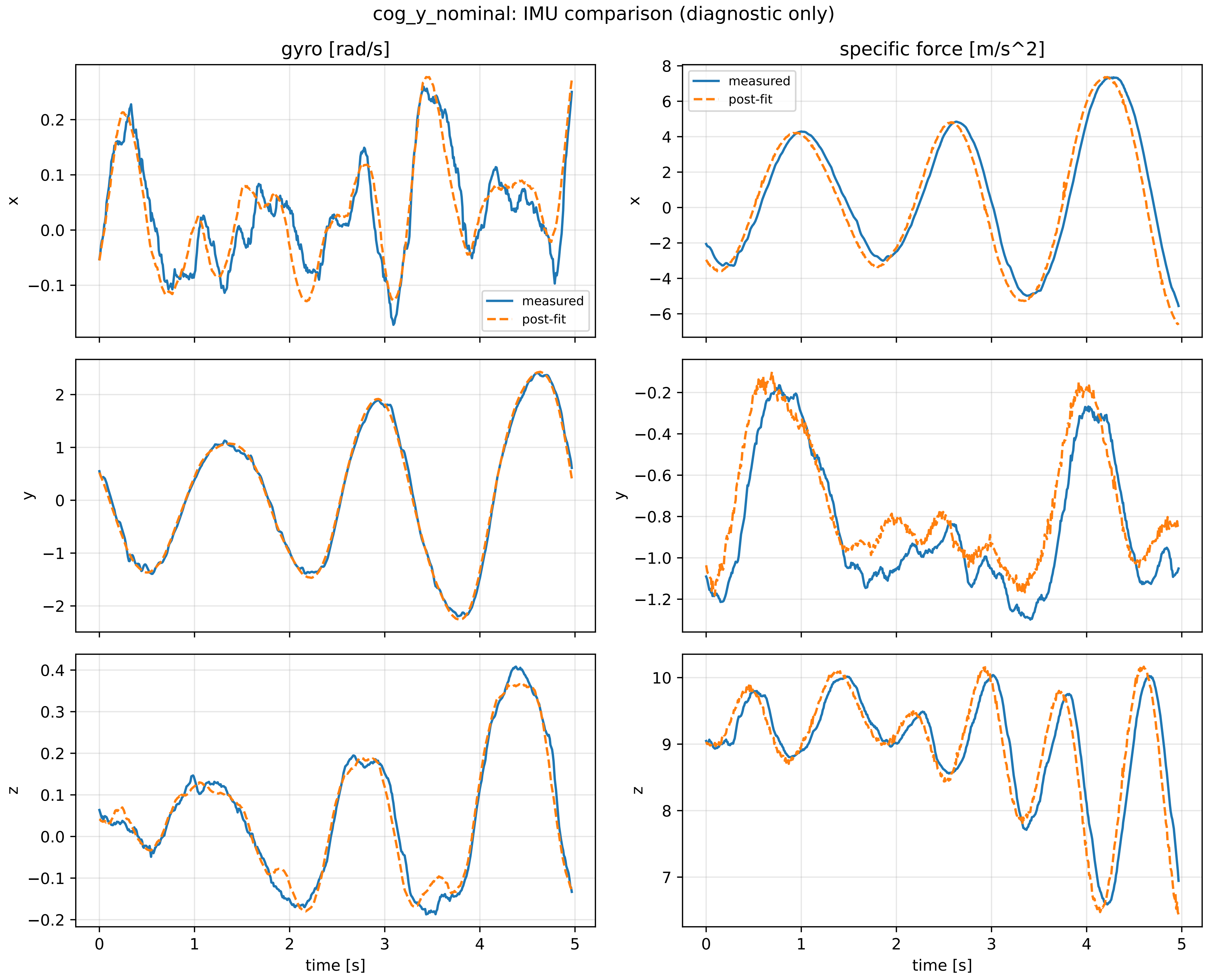
cog_y_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.009 0.016 0.035 0.26 0.739]

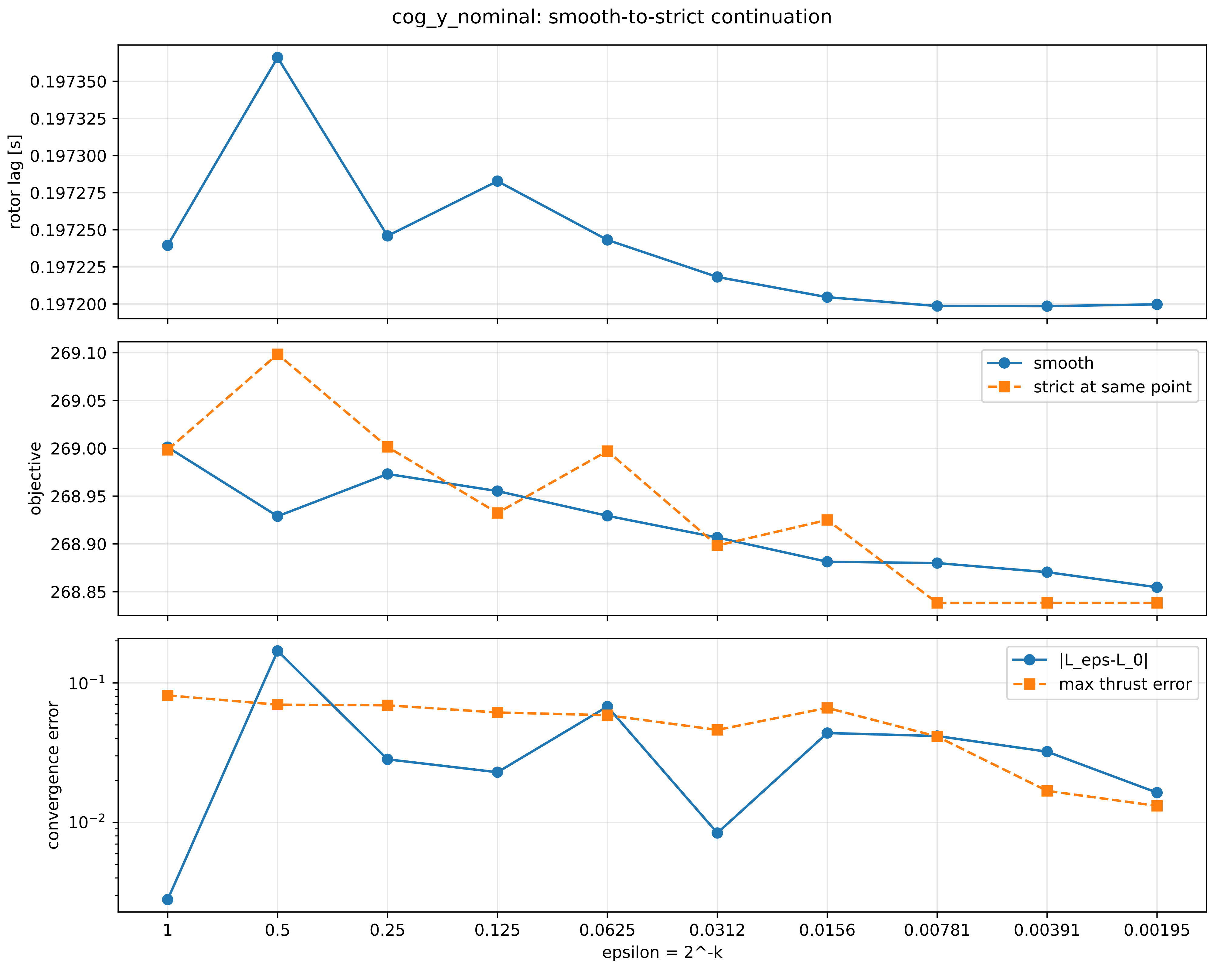




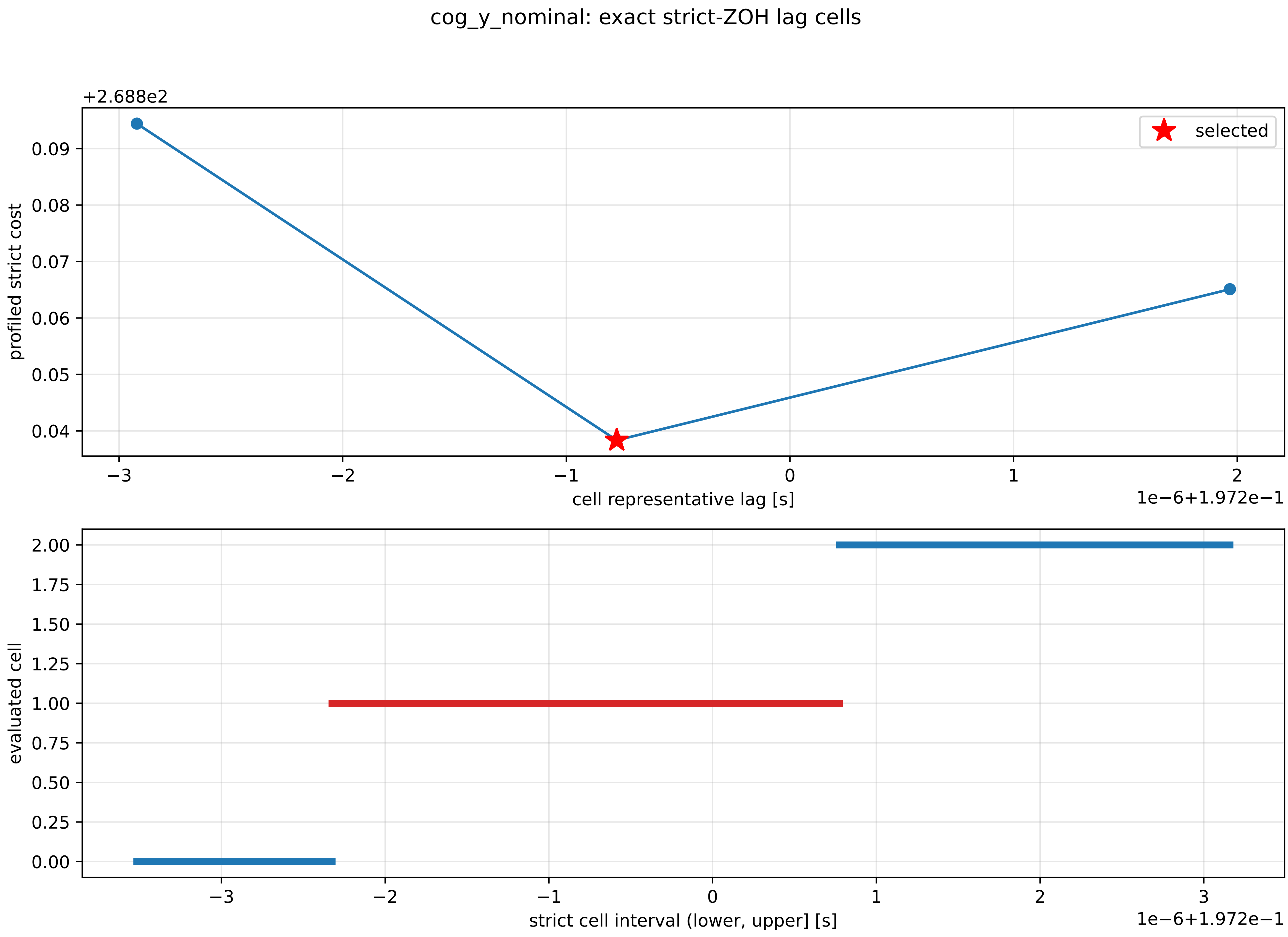
cog_y_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]


cog_y_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.0759, max=0.2095 rad
gimbal 2: RMSE=0.08099, max=0.2013 rad
gimbal 3: RMSE=0.07393, max=0.172 rad
gimbal 4: RMSE=0.07131, max=0.1737 rad
raw
SG
command replay
angle [rad]
time [s]


cog_y_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



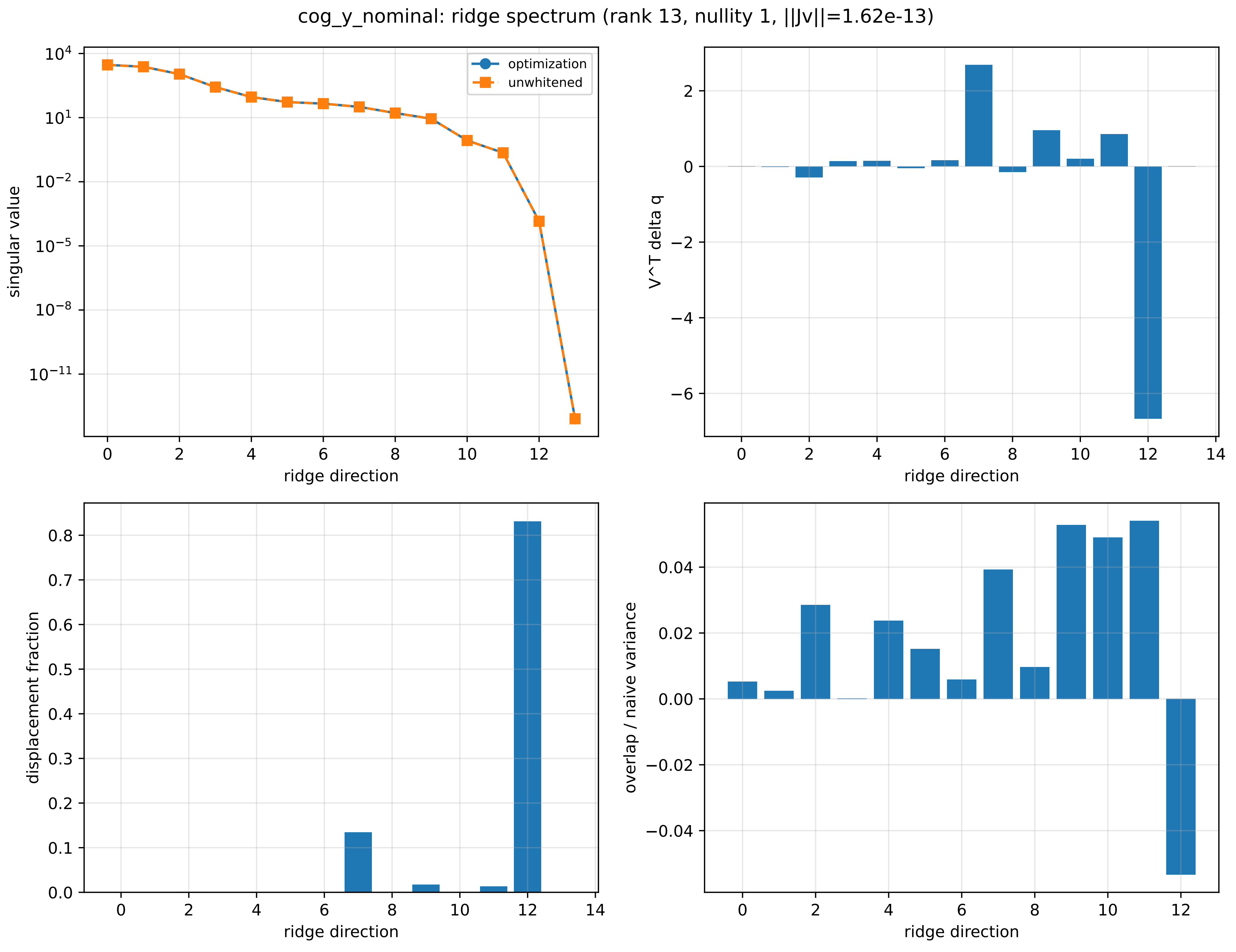
cog_y_nominal: smooth-to-strict continuation
rotor lag [s]
0.197350
0.197325
0.197300
0.197275
0.197250
0.197225
0.197200
objective
269.10
269.05
269.00
268.95
268.90
268.85
smooth
strict at same point
convergence error
10^{-1}
10^{-2}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k



selected (0.197197676, 0.197200775] s; final smooth 0.197199765 s; data boundary=False

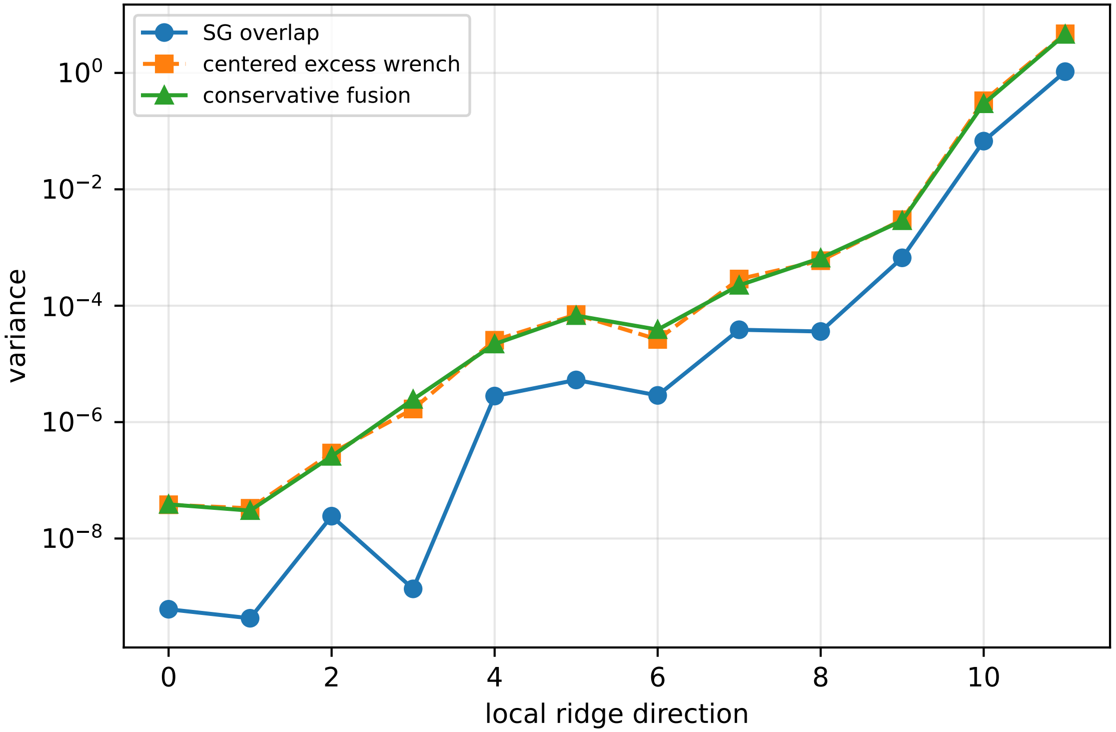
cog_y_nominal: parameter estimate

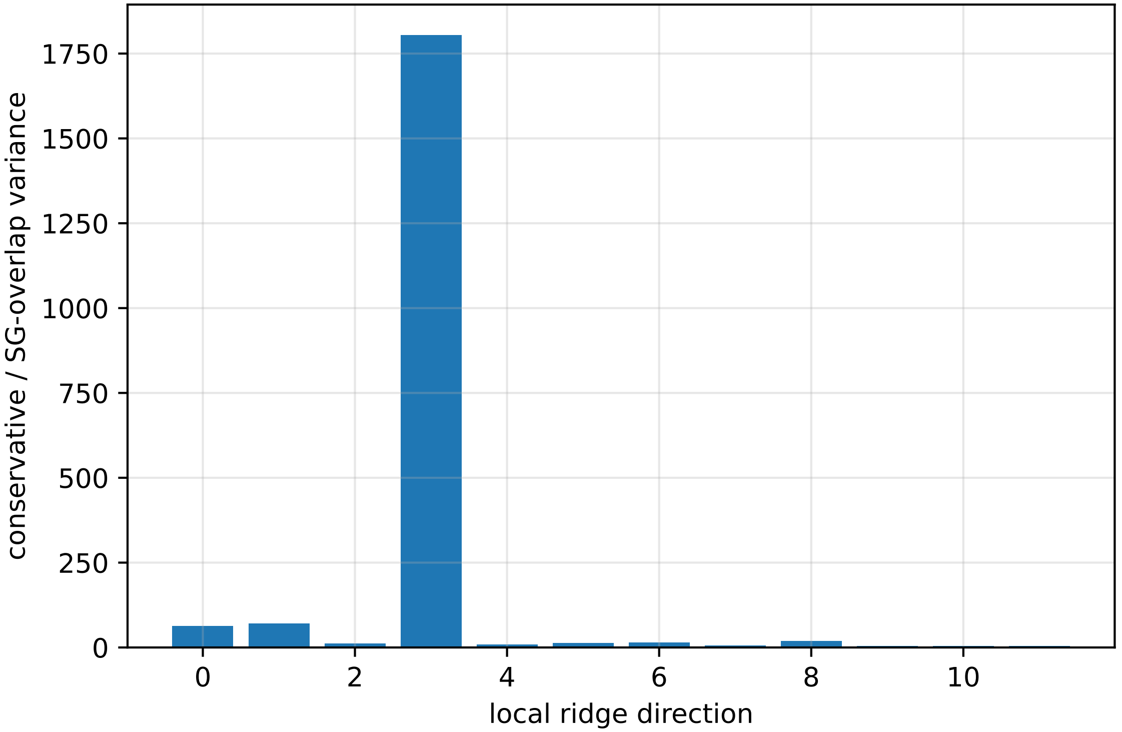
```
Case: cog_y_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.41406167
inertia [kg m^2]:
[[ 3.37278101e-01 -8.52387738e-05 -5.63167124e-02]
 [-8.52387738e-05  2.65405718e-01 -5.74761683e-03]
 [-5.63167124e-02 -5.74761683e-03  5.77370561e-01]]
CoG body [m]: [ 1.54815021e-02 -3.05413590e-05 -4.45368332e-02]
force effectiveness: [1.13611044 1.43796218 1.29508404 1.09604879]

scale-free J/m [m^2]:
[[ 9.87908636e-02 -2.49669695e-05 -1.64955170e-02]
 [-2.49669695e-05  7.77389935e-02 -1.68351289e-03]
 [-1.64955170e-02 -1.68351289e-03  1.69115446e-01]]
scale-free f/m: [0.33277385 0.42118811 0.37933821 0.32103954]

rotor lag cell: (0.197197676, 0.197200775] s
representative: 0.197199225 s
```

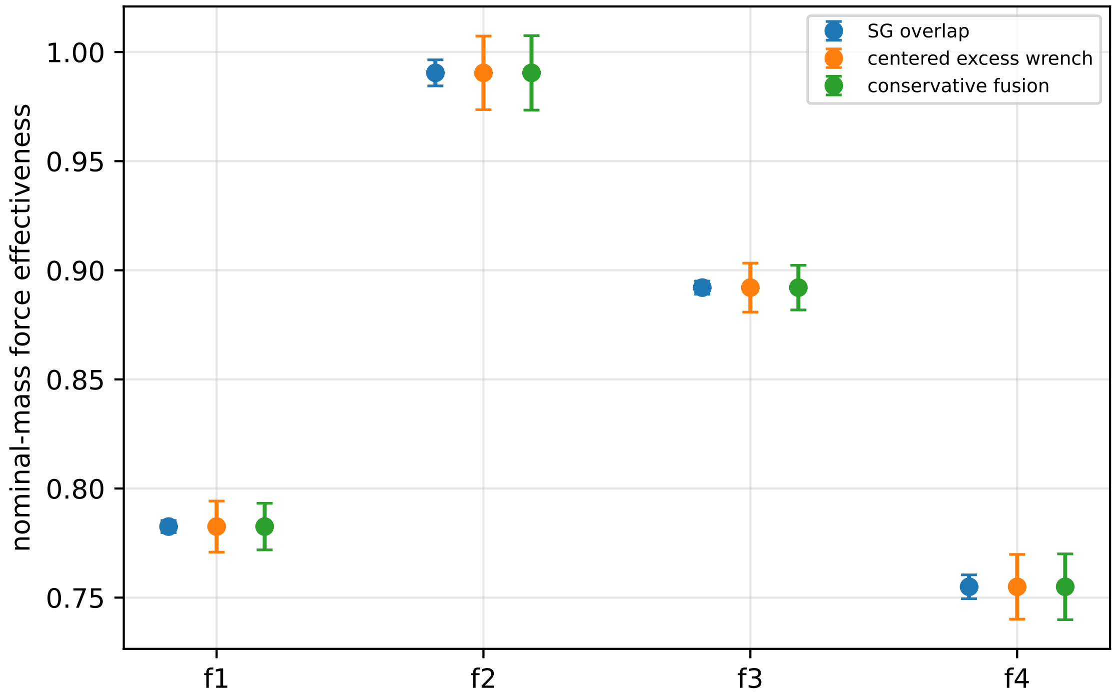


cog_y_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.62e-13)
singular value
optimization
unwhitened
ridge direction
V^T delta q
displacement fraction
overlap / naive variance

cog_y_nominal: Conservative fusion uncertainty

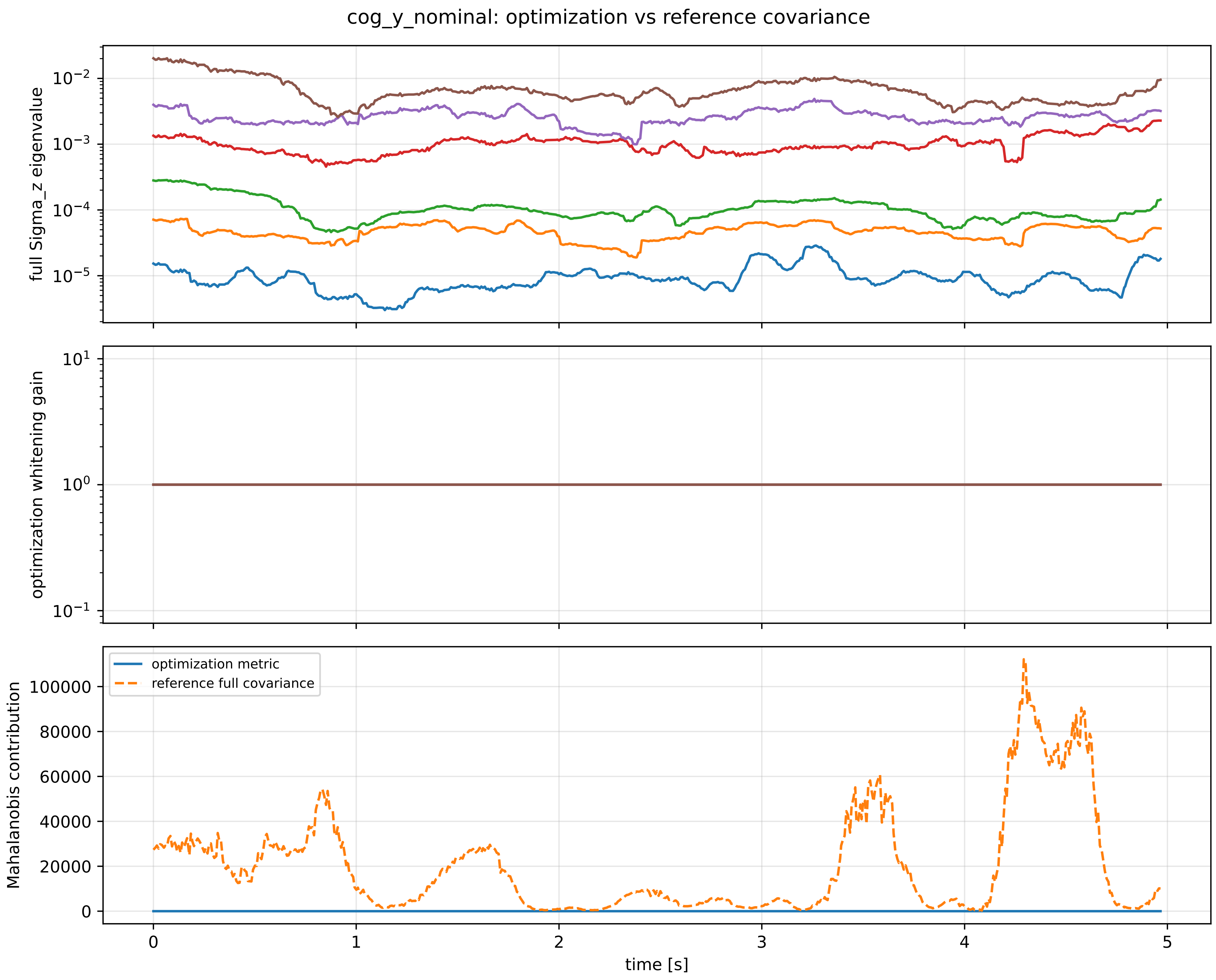




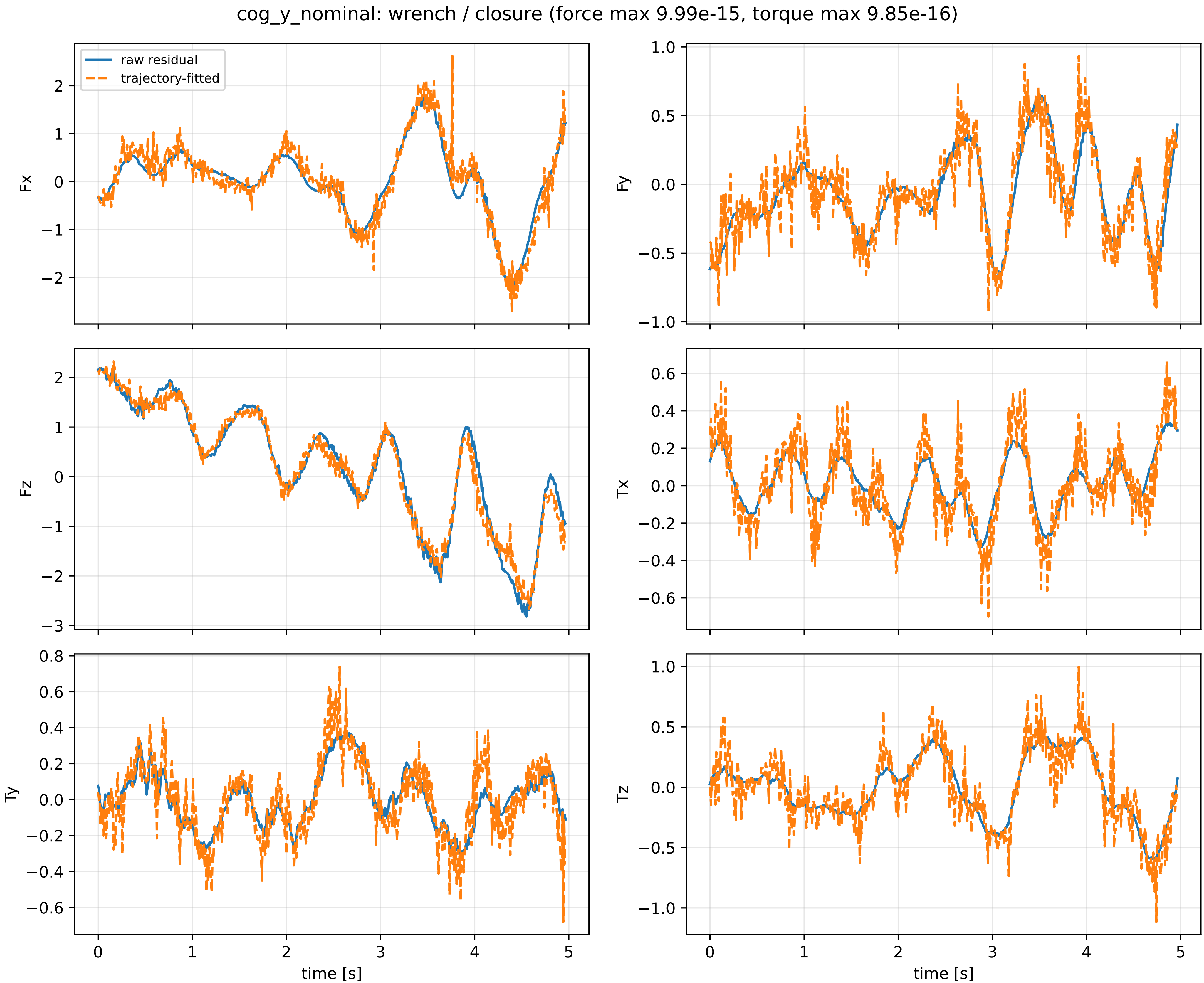



```
Top three non-gauge ambiguous directions
tr(M_res) / tr(M_SG) = 44.59
wrench-to-acceleration recovery max error = 3.77e-15
exact scale gauge ridge index = 13 (alignment 1)

index 11: variance=4.594, ratio=4.366
  log_mass=-0.0474; second_moment_diag_1=-0.424
  second_moment_diag_2=-0.0252; second_moment_diag_3=+0.686
  second_moment_offdiag_12=+0.154; second_moment_offdiag_13=-0.54
  second_moment_offdiag_23=-0.151; cog_x=+2.72e-06
  cog_y=-0.000209; cog_z=+2.3e-05
  log_force_eff_1=-0.0468; log_force_eff_2=-0.0464
  log_force_eff_3=-0.0485; log_force_eff_4=-0.0479

index 10: variance=0.2953, ratio=4.388
  log_mass=-0.0102; second_moment_diag_1=+0.00626
  second_moment_diag_2=-0.122; second_moment_diag_3=+0.163
  second_moment_offdiag_12=-0.86; second_moment_offdiag_13=-0.155
  second_moment_offdiag_23=+0.439; cog_x=-0.00148
  cog_y=+0.00356; cog_z=-0.000378
  log_force_eff_1=-0.0181; log_force_eff_2=-0.0288
  log_force_eff_3=-0.00292; log_force_eff_4=+0.0122

index 9: variance=0.002889, ratio=4.344
  log_mass=-0.153; second_moment_diag_1=-0.0294
  second_moment_diag_2=+0.896; second_moment_diag_3=-0.0694
  second_moment_offdiag_12=-0.0706; second_moment_offdiag_13=-0.0847
  second_moment_offdiag_23=+0.0995; cog_x=+0.0205
  cog_y=-0.00981; cog_z=-0.00478
  log_force_eff_1=-0.2; log_force_eff_2=-0.0446
  log_force_eff_3=-0.0912; log_force_eff_4=-0.308
```

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.



cog_y_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
10^{-2}
10^{-3}
10^{-4}
10^{-5}
optimization whitening gain
10^{1}
10^{0}
10^{-1}
Mahalanobis contribution
100000
80000
60000
40000
20000
0
optimization metric
reference full covariance
0
1
2
3
4
5
time [s]



cog_y_nominal: wrench / closure (force max 9.99e-15, torque max 9.85e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

cog_y_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_cog_y_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/cog_y_nominal.j
SHA256: 3efc6d36ba4fff0dc12548d07dd00c0a5c4bcb02e99decb13ffa2adae4ce3f12

data objective:  268.838338689
prior objective: 1.09224633169e-06
total objective: 268.838339782

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: cog_y_nominal (cog_position_body_m)
  components: y
  target: [-3.05265789e-05]
  std:    [1.e-05]
  final:  [-3.0541359e-05]
  error:  [-1.47800293e-08]
  standardized residual: [-0.001478]
  factor objective: 1.09224633169e-06
```